

INFO 510 Bayesian Modelling and Inference

Lecture 14 – Gibbs Sampling (MCMC algorithm)

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November 17, 2025

Markov Chain

Markov chain is a process that consists of a finite number of states and some known probabilities $\mathbf{p}_{ij}$, where $\mathbf{p}_{ij}$ is the probability of moving from state i to state j .

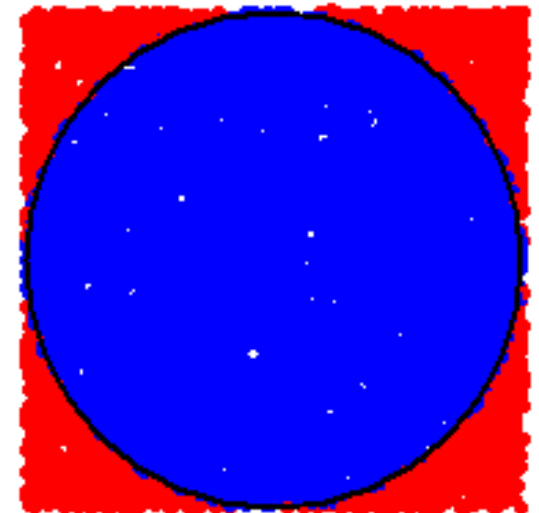
Transition probability (transition matrix)

$$P(X_{t+1} = s \mid X_t = s_t, \cancel{X_{t-1} = s_{t-1}}, \dots, \cancel{X_0 = s_0}) = P(X_{t+1} = s \mid X_t = s_t)$$

distribution of X_{t+1}
depends on X_t
whatever happened before time t doesn't matter.

Monte Carlo methods

Using randomization to reach to a solution where an exact solution is impossible



Metropolis-Hastings Algorithm

Steps of the Algorithm

Initialize - start with an arbitrary initial point x_t

Propose - from the current state x_t , propose a new state x' using a proposal distribution $q(x'|x_t)$

Compute the acceptance probability

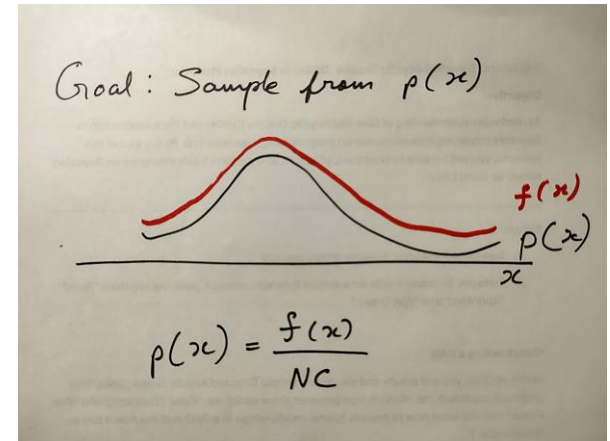
$$\alpha = \min \left(1, \frac{\pi(x')q(x_t|x')}{\pi(x_t)q(x'|x_t)} \right)$$

Accept or Reject -

With probability α , accept the new state x' (set $x_{t+1}=x'$)

Otherwise, stay at the current state $x_{t+1} = x_t$

Iterate this process to generate a chain of samples

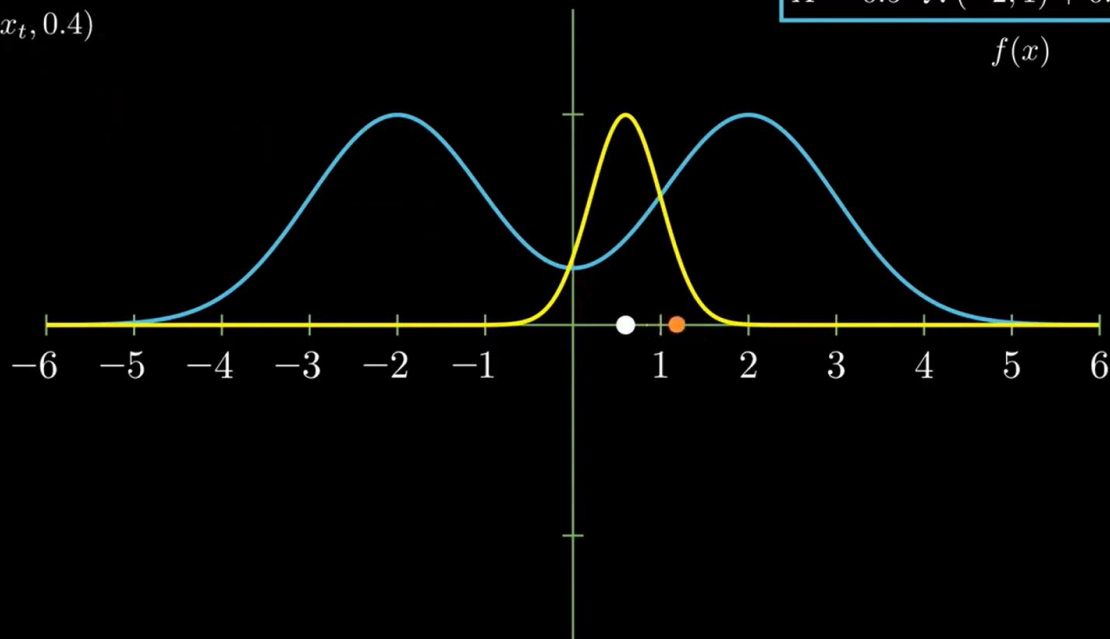


$$x_t = 0.6$$

$$x_{t+1} \sim \mathcal{N}(x_t, 0.4)$$

Target Distribution

$$X \sim 0.5 \cdot \mathcal{N}(-2, 1) + 0.5 \cdot \mathcal{N}(2, 1)$$

 $f(x)$ 

Metropolis-Hastings Algorithm

Limitations

The choice of proposal distribution $q(x'|x)$ critically affects performance

MH often exhibits random walk behavior, particularly in high dimensions
Slow exploration of parameter space, high autocorrelation

Every proposed move requires computing the acceptance probability

Rejected proposals waste computation while the chain stays in place

High rejection rates mean longer burn-in periods

If correlated variables, inefficient sampling

Gibbs sampling

No Proposal Distribution Needed

Gibbs sampling draws from the exact conditional distributions; there are no rejections, every sample is accepted

By conditioning on all other variables, Gibbs sampling naturally adapts to the correlation structure in the target distribution

straightforward to implement where conditional distributions have standard forms

Gibbs Sampling

sample from the conditional distributions of the variables, one at a time, in a cyclic manner

well-suited when the joint distribution is difficult to sample from directly, but the conditional distributions of each variable are tractable

given a multivariate distribution, it's simpler to sample from a conditional distribution than from a joint distribution

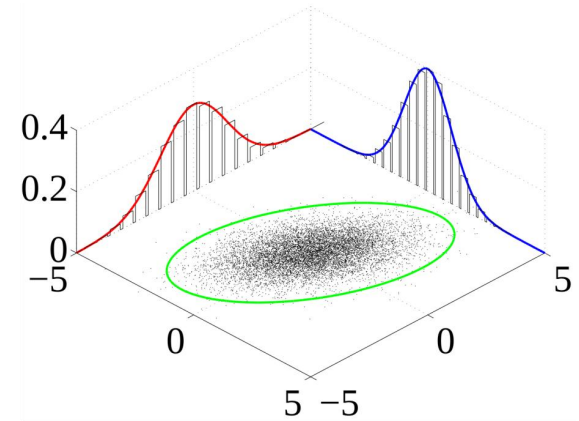
$P(x,y)$ or $f(x,y)$ – difficult to sample

Instead, $P(x|y)$ and $P(y|x)$ – Easy to sample

Joint probability distribution

Two random variables X and Y

The joint probability distribution is a function that assigns probabilities to every possible pair of values that X and Y can take



Example:

Relationship between the weather and traffic conditions

Random Variable 1 (X): Whether it is raining (yes = 1, no = 0)

Random Variable 2 (Y): Whether the traffic is heavy (yes = 1, no = 0)

Example:

Relationship between the weather and traffic conditions

Random Variable 1 (X): Whether it is raining (yes = 1, no = 0)

Random Variable 2 (Y): Whether the traffic is heavy (yes = 1, no = 0)

$P(\text{Rain} = 1, \text{Traffic} = 1) = 0.4$

This means there's a 40% chance that it's both raining and there is heavy traffic

$P(\text{Rain} = 0, \text{Traffic} = 1) = 0.2$

There's a 20% chance that it's not raining, but traffic is still heavy

$P(\text{Rain} = 1, \text{Traffic} = 0) = 0.1$

There's a 10% chance it's raining, but traffic is light

$P(\text{Rain} = 0, \text{Traffic} = 0) = 0.3$

There's a 30% chance it's not raining, and traffic is light

The sum of all these probabilities must equal 1 because one of these situations must happen

A joint probability distribution is also the foundation for two important concepts – marginal probability and conditional probability

Marginal Probability - This is the probability of one variable, ignoring the other

$$\begin{aligned} P(\text{Rain}) &= P(\text{Rain} = 1, \text{Traffic} = 1) + P(\text{Rain} = 1, \text{Traffic} = 0) \\ &= 0.4 + 0.1 = 0.5 \end{aligned}$$

Conditional Probability – This is the probability of one variable given that another variable has occurred

$$P(\text{Rain} = 1 \mid \text{Traffic} = 1) = P(\text{Rain} = 1 \text{ and Traffic} = 1) \div P(\text{Traffic} = 1)$$

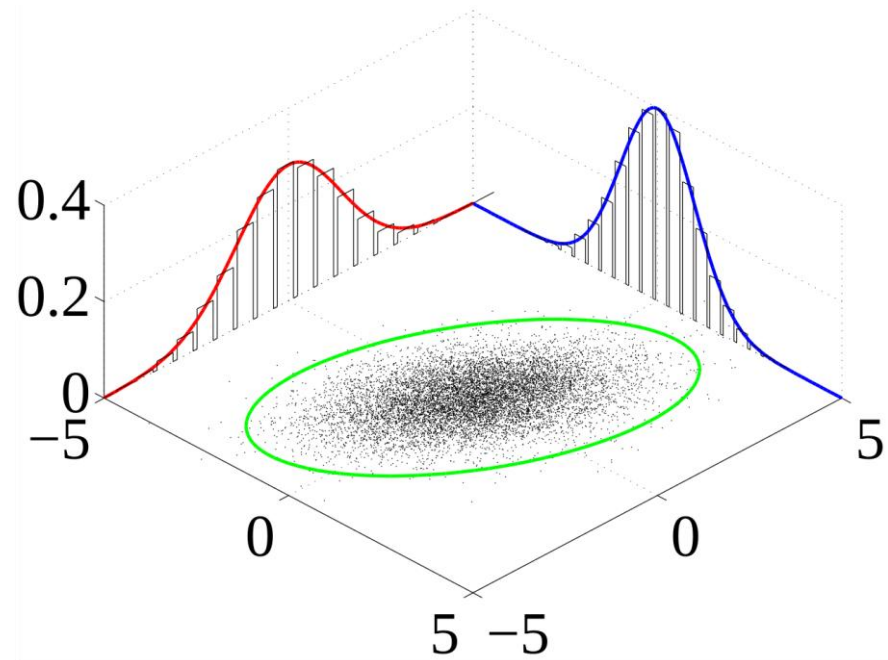
Given that traffic is heavy, how likely is it that it's also raining?

Joint Probability Distribution for Continuous Random Variables

Represented by a joint probability density function $f(x,y)$, which gives the probability density at any point (x,y) in the space of the two variables $\int$

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x,y) dy dx$$

This integral gives the probability that X lies between a and b and Y lies between c and d



Gibbs Sampling

The Gibbs Sampling Algorithm

Suppose $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ is our parameter vector.

Initialize $\theta^{(0)} = (\theta_1^{(0)}, \theta_2^{(0)}, \dots, \theta_p^{(0)})$

For iterations $t = 1, 2, \dots, T$:

Sample $\theta_1^{(t)} \sim p(\theta_1, \mid \theta_2^{(t-1)}, \theta_3^{(t-1)}, \dots, \theta_p^{(t-1)} y)$

Sample $\theta_2^{(t)} \sim p(\theta_2, \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \dots, \theta_p^{(t-1)} y)$

Sample $\theta_3^{(t)} \sim p(\theta_3, \mid \theta_1^{(t)}, \theta_2^{(t)}, \dots, \theta_p^{(t-1)} y) \dots P$

Sample $\theta_p^{(t)} \sim p(\theta_p, \mid \theta_1^{(t)}, \theta_2^{(t)}, \dots, \theta_{p-1}^{(t)} y)$

We immediately use updated values within the same iteration.

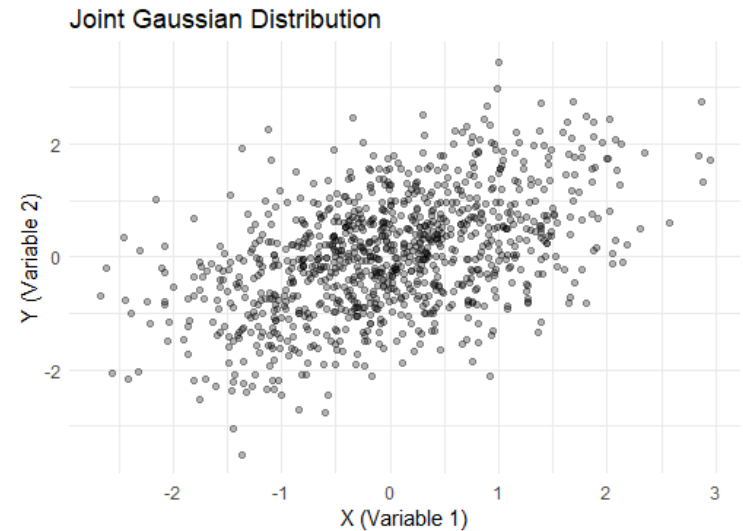
Gibbs Sampling

Allows you to sample from conditional distribution of a variable, in a multi variate distribution, keeping other variables fixed

Normal distribution of 2 random variables
x and y with $\mu = 0$

Co-variance
matrix

$$\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$



$P(x,y)$ or $f(x,y)$ – difficult to sample

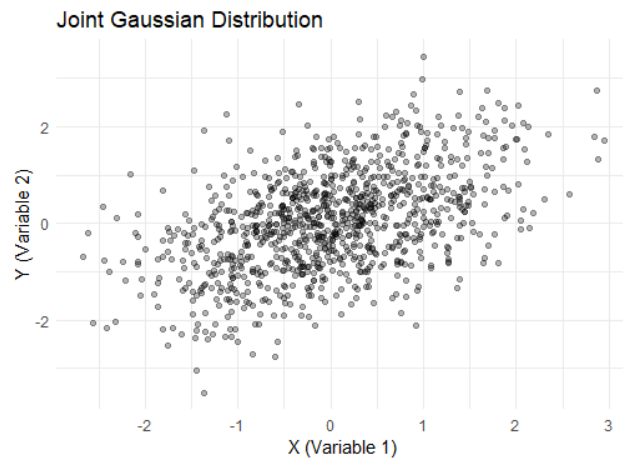
Instead, $P(x|y)$ and $P(y|x)$ – Easy to sample

$$P(x|y) = N(\rho y, 1 - \rho^2) \\ = N(y/2, 3/4)$$

$$P(y|x) = N(\rho x, 1 - \rho^2) \\ = N(x/2, 3/4)$$

* ρ = correlation between x and y

If you have a fixed value of y and you want to sample x, then you can sample from the single variable normal distribution with mean y/2 and variance of 3/4



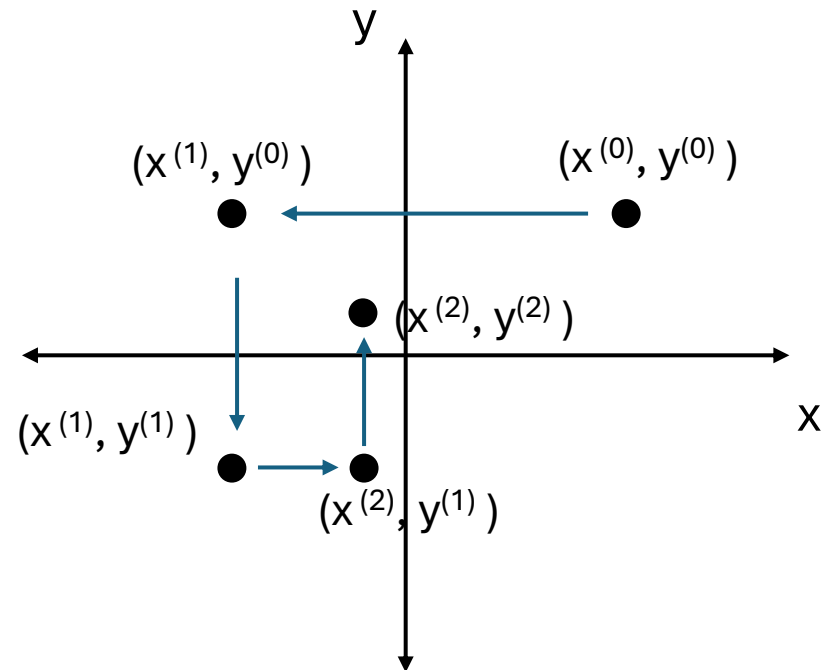
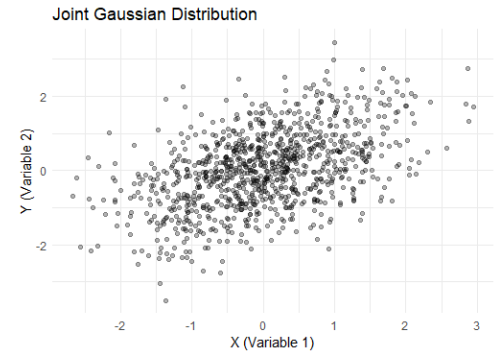
Gibbs Sampling procedure

Start at $x^{(0)}, y^{(0)}$

Sample $x^{(1)} \sim P(x^{(1)} | y^{(0)})$

Sample $y^{(1)} \sim P(y^{(1)} | x^{(1)})$

Sample $x^{(2)} \sim P(x^{(2)} | y^{(1)})$



Keep alternating between x and y , each time sampling from their conditional distributions.

Eventually, the sequence of samples will converge to the true joint distribution.

Mixed Joint Distributions

consider a case where one variable is discrete, and the other is continuous

$$P(Y|X=0) \sim N(\mu_0, \sigma_0^2)$$

X is discrete binary variable

$$P(Y|X=1) \sim N(\mu_1, \sigma_1^2)$$

Y is continuous variable

$P(X|Y)$ is a Bernoulli distribution depending on Y

Start with some initial values for X and Y

Sample X from $P(X|Y)$ – Bernoulli

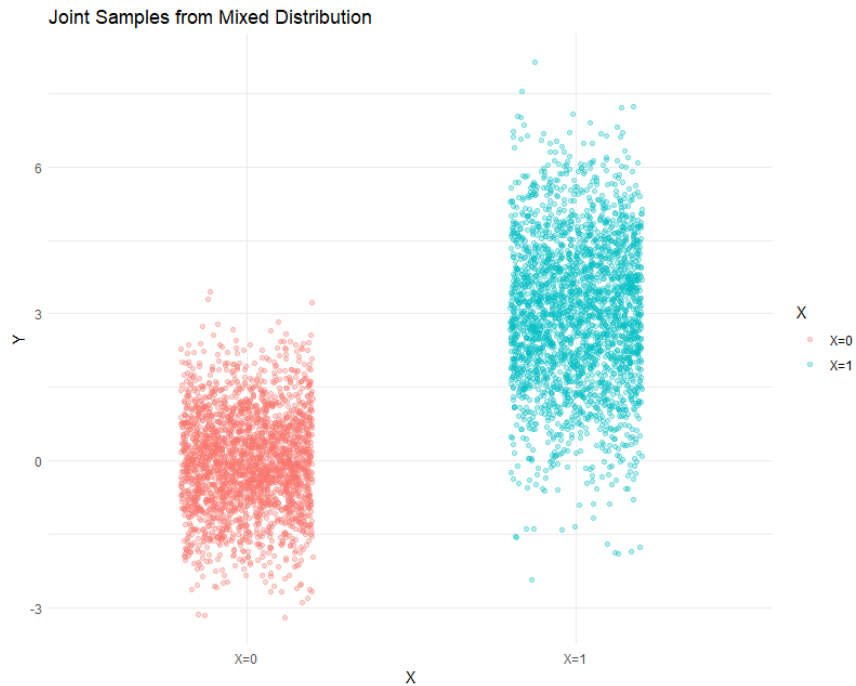
distribution

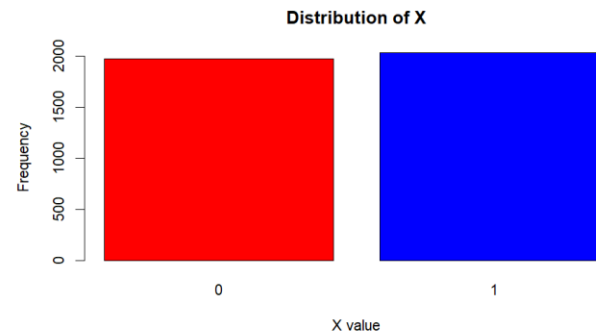
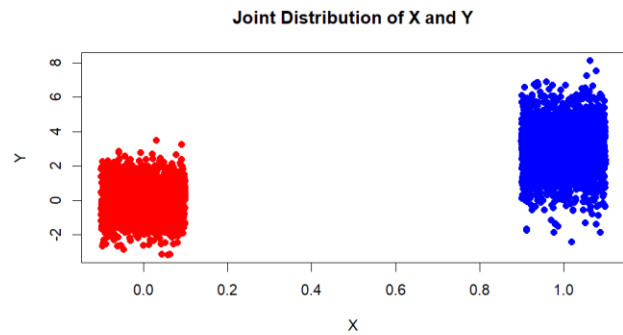
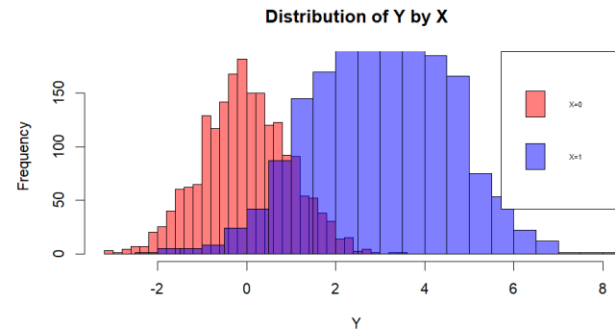
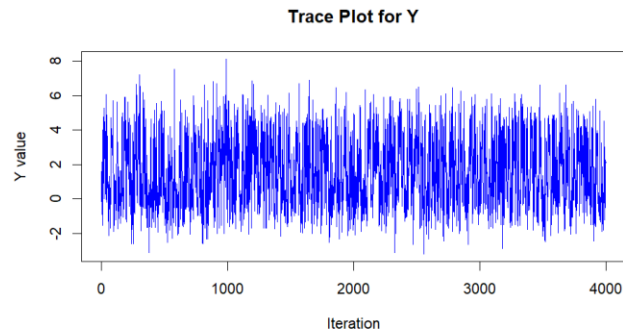
$$P(X = 1|Y) = \frac{P(X = 1)P(Y|X = 1)}{P(Y)} = \frac{P(X = 1)\mathcal{N}(Y; \mu_1, \sigma_1^2)}{P(Y)}$$

Sample Y from $P(Y|X)$ – Depending on whether $X=0$ or $X=1$, sample Y from the corresponding normal distribution

Repeat – alternate between sampling X and Y

Mixed Joint Distributions





μ of $Y|X=0$: -0.02237872

μ of $Y|X=1$: 3.011483

Gibbs Sampling for the Beta-Binomial Model

Mixed distribution, where the parameter of a Binomial distribution is drawn from a Beta distribution

Binomial distribution models discrete outcomes (such as the number of successes in a series of trials), while the Beta distribution models the unknown probability of success; e.g. coin toss

Let's say we have two variables

p , the probability of success in a binomial trial, follows a Beta distribution $p \sim \text{Beta}(\alpha, \beta)$

n , the number of successes in N trials, follows a Binomial distribution: $n \sim \text{Binomial}(N, p)$

We want to sample from the joint distribution $P(p, n)$ which is a mixture of discrete and continuous components

Gibbs Sampling for the Beta-Binomial Model

Gibbs Sampling Process

Initialize p and n

Start with an initial guess for the success probability p and the number of successes n

Sample n from $P(n|p)$

Given p , the number of successes n is drawn from a Binomial distribution - $n \sim \text{Binomial}(N, p)$

Sample a new value for n , say $n=5$ if $N=10$ and $p=0.5$

Sample p from $P(p|n)$

Given n , the success probability p is drawn from a Beta posterior distribution $p \sim \text{Beta}(\alpha+n, \beta+N-n)$

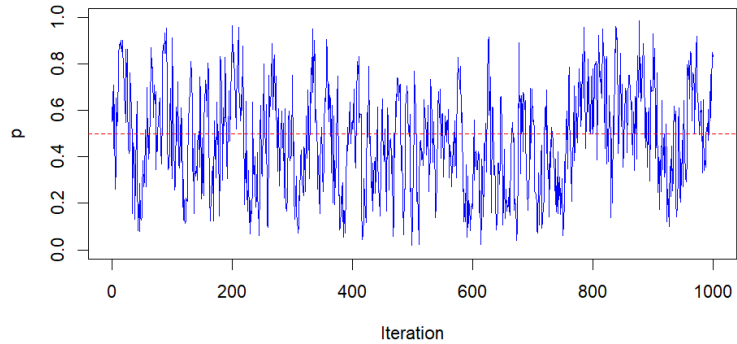
Sample a new value for p , say $p=0.6$

Repeat - Keep alternating between sampling n and p

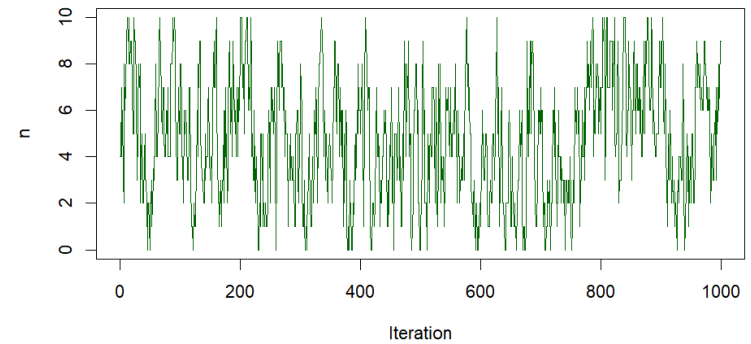
Over time, the samples will converge to the joint distribution $P(p, n)$

Gibbs Sampling for the Beta-Binomial Model

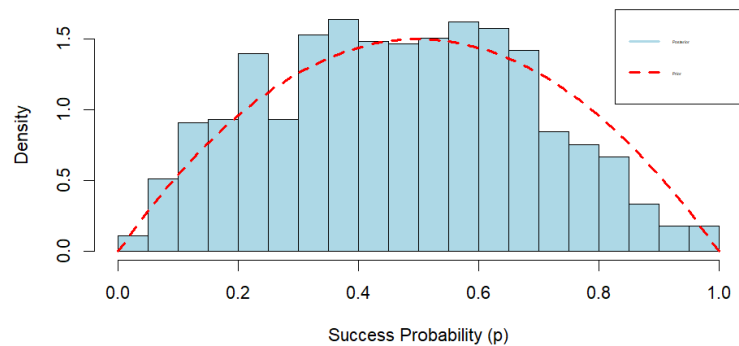
Trace Plot: Success Probability (p)



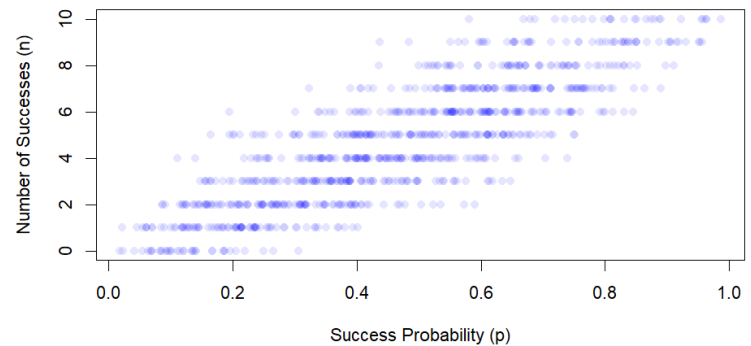
Trace Plot: Number of Successes (n)



Posterior Distribution of p



Joint Distribution $P(p, n)$



Gibbs Sampling for Dirichlet-Multinomial Models

Instead of dealing with a binary outcome (success or failure), the Dirichlet-Multinomial model deals with categorical outcomes across multiple categories

we are interested in modeling the following joint distribution;

X , a set of counts for each category, follows a Multinomial distribution - $X \sim \text{Multinomial}(N, \theta)$

θ , the vector of probabilities for each category, follows a Dirichlet distribution - $\theta \sim \text{Dirichlet}(\alpha)$

Gibbs Sampling Process

Initialize X and θ

Start with initial guesses for the category counts $X=(x_1, x_2, \dots, x_k)$ and the probabilities $\theta = (\theta_1, \theta_2, \dots, \theta_k)$

Sample X from $P(X|\theta)$

Given θ , sample the category counts X from the Multinomial distribution:

$$X \sim \text{Multinomial}(N, \theta)$$

This means we draw a new set of counts for each category based on the current probabilities θ

Sample θ from $P(\theta|X)$

Given X , sample the category probabilities θ from a Dirichlet posterior distribution

$$\theta \sim \text{Dirichlet}(\alpha_1 + x_1, \alpha_2 + x_2, \dots, \alpha_k + x_k)$$

Sample a new vector $\theta = (\theta_1, \theta_2, \dots, \theta_k)$

Repeat

Continue alternating between sampling X and θ

Over time, the samples will converge to the joint distribution $P(X, \theta)$

Gibbs Sampling

Advantages

Simplicity - Only requires the ability to sample from conditional distributions

Efficiency - Works well for high-dimensional problems where the joint distribution is difficult to sample directly

Flexibility - Can be applied to a wide range of models in Bayesian inference

Limitations

Slow Convergence - In some cases, the algorithm may take many iterations to converge

Inapplicable in Non-Tractable Conditionals - If the conditional distributions are not easy to compute or sample from, Gibbs Sampling is not suitable

Applications of Gibbs Sampling

Bayesian Hierarchical Models - When the joint posterior is too complex to sample from directly, Gibbs Sampling helps break the problem into simpler conditional distributions

Latent Variable Models - Gibbs Sampling is used to sample latent variables, making it applicable in models like Hidden Markov Models (HMMs)

Gaussian Mixture Models - In mixture models, Gibbs Sampling is used to estimate the posterior distributions of parameters

***Gibbs sampling is more structured than Metropolis-Hastings. It works by splitting the problem into smaller, manageable pieces and handling them one at a time. The method will always accept the new sample since it's directly sampling from the conditional distributions, not proposing candidates that could be rejected*